COMPUTER-AIDED DRUG DESIGN

Parkinson's Disease Biological Target: LRRK2 (Leucine-Rich Repeat Kinase 2)

PROJECT TEAM

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1. Introduction

1.1 Understanding Parkinson's Disease

Only Alzheimer's disease is more common than Parkinson's disease (PD), which is the second most common neurodegenerative illness worldwide. An estimated 10 million people worldwide are affected by it as of 2024, and the incidence is quickly increasing in those over 60. Compared to other prevalent diseases, Parkinson's disease is still poorly understood at the molecular level despite its widespread impact. More importantly, there is currently neither a cure nor a treatment that has been shown to slow or stop the underlying neurodegeneration.

Parkinson's disease is fundamentally a movement problem. The progressive, targeted death of dopaminergic neurons in the substantia nigra pars compacta, a tiny but crucial area of the midbrain, is the characteristic pathogenic event.

These neurons are responsible for producing dopamine — a neurotransmitter that plays a central role in coordinating smooth, controlled body movements. As these neurons die, dopamine levels in the brain fall dramatically, and the resulting neurotransmitter imbalance disrupts the basal ganglia circuit that governs voluntary movement. The clinical consequence of this is the characteristic motor triad of PD: resting tremor (involuntary shaking at rest), rigidity (stiffness and resistance in limbs), and bradykinesia (slowed, reduced movement). As the disease progresses, patients also develop postural instability, freezing of gait, and in later stages, dementia, depression, and autonomic dysfunction. The death of dopaminergic neurons in PD is not random — it is closely associated with the intracellular accumulation of misfolded alpha-synuclein protein into pathological aggregates known as Lewy bodies.

The neuropathological characteristic of Parkinson's disease is these protein deposits. For the past thirty years, one of the main issues in PD research has been figuring out what causes them to form and what causes the subsequent neuronal death.

There are currently only symptomatic treatments available for Parkinson's disease. After more than 50 years, levodopa (L-DOPA), a dopamine precursor that may pass across the blood-brain barrier, is still the most effective treatment for symptoms. To treat motor symptoms, a combination of dopamine agonists (Pramipexole, Ropinirole, Rotigotine), MAO-B inhibitors (Selegiline, Rasagiline, Safinamide), and COMT inhibitors (Entacapone) is employed. On-off fluctuations, dyskinesias, and wearing-off are among the motor problems that are inevitably brought on by extended treatment. The underlying neurodegeneration is not stopped by any of these medications. This unmet therapeutic need drives intense interest in identifying disease-modifying targets, proteins whose inhibition might slow or stop neuronal death rather than simply compensating for it.

1.2 The Biological Justification for LRRK2

LRRK2 (Leucine-Rich Repeat Kinase 2) is likely the most pharmacologically tractable target among the numerous proteins linked to Parkinson's disease. LRRK2 is a large, multifunctional kinase that has multiple protein-protein interaction domains that mediate its scaffolding capabilities in addition to a kinase domain that transfers phosphate groups to target proteins and a GTPase domain that binds and hydrolyzes GTP. Because of its complexity, LRRK2 is a key player in several intracellular signaling pathways, such as lysosomal function, autophagy, mitophagy, and vesicle trafficking.

The finding that gain-of-function mutations in the LRRK2 gene are the most prevalent known monogenic cause of both familial and sporadic Parkinson's disease made the clinical significance of LRRK2 abundantly evident. Particularly common is the G2019S mutation in the kinase domain, which accounts for 30–40% of North African Arab populations, up to 13% of Ashkenazi Jewish populations, and 1-2% of sporadic PD cases in European populations. Endosomal and lysosomal trafficking are disrupted by the G2019S mutation, which raises LRRK2 kinase activity by about two to three times. This causes hyperphosphorylation of downstream substrates, most notably the Rab GTPase family. It is believed that the accumulation of misfolded proteins and eventual neuronal death are caused by the ensuing malfunction in cellular quality control mechanisms.

Importantly, G2019S LRRK2's aberrantly high kinase activity is its pathogenic mechanism. Kinase inhibition is therefore a simple treatment hypothesis: if hyperactive LRRK2 kills neurons, then bringing its kinase activity back to normal ought to be neuroprotective. LRRK2 kinase inhibitors regularly lower Rab phosphorylation, shield neurons from LRRK2-driven toxicity, and restore autophagy deficiencies, all of which have been thoroughly demonstrated in cellular and animal models. A number of LRRK2

inhibitors have started clinical trials, and there is a clear scientific path from this computational work to the eventual creation of a treatment.

1.3 The Case for Computational Drug Discovery

Finding a therapeutically useful target, like LRRK2, is just the beginning. Finding a little chemical that penetrates the brain, binds to it firmly and selectively, and is safe enough for long-term human use is the more difficult task. High-throughput screening (HTS), which literally tests hundreds of thousands of chemical compounds in biochemical assays to identify those that block the target, is how traditional drug discovery tackles this issue. Although HTS is quite strong, it is also very costly, sluggish, and resource-intensive. Nowadays, it costs more than USD 2.6 billion on average to bring a new medication to market, and it takes 12 to 15 years from target identification to approval. In late clinical stages, when failure is most expensive, the great majority of drug candidates fail.

One revolutionary solution to these inefficiencies is Computer-Aided Drug Design (CADD). CADD significantly reduces the number of compounds that must be synthesized and evaluated by using computational and statistical techniques to solve the drug discovery problem. This allows researchers to intelligently prioritize compounds before any chemistry is done in the laboratory. This initiative focuses on ligand-based CADD (LBCADD), wherein mathematical models that can predict the affinity of novel, untested compounds are trained using the chemical structures and measured binding affinities of existing compounds. Ligand-based procedures are accessible, scalable, and especially useful for early-stage drug development because they can function just from chemical and bioactivity data, in contrast to structure-based methods that need a high-resolution crystal structure of the target protein.
This study focuses on quantitative structure-activity relationship (QSAR) modeling for LRRK2 inhibition. Can a machine learning algorithm correctly predict a molecule's IC50 value against LRRK2 given its chemical structure (recorded as a numerical feature vector)? If so, the trained model turns into a virtual screening tool that helps researchers choose the most promising candidates for experimental testing by ranking millions of commercially accessible chemicals based on their anticipated potency. This is more than just a computational exercise; it is a true scientific contribution to Parkinson's disease treatment research.

2.PROJECT OBJECTIVE

Data science, cheminformatics, machine learning, and pharmacological validation were all part of the project's well-defined goals. A logical end-to-end drug discovery pipeline is formed by each goal building on the one before it.
• Beginning with over 23,000 raw hits, retrieve and curate LRRK2-specific bioactivity data (IC50 values, nM) from BindingDB, the biggest open library of measured protein-ligand binding affinities in the world.• To clean, standardize, and convert IC50 values onto the logarithmic pIC50 scale—the favored target variable for QSAR regression because of its linear relationship with binding free energy—use a strict multi-step data pretreatment workflow.
• Create two complimentary representations of chemical features: (1) continuous PaDEL molecular descriptors that encode physicochemical and topological aspects, and (2) binary Morgan fingerprints that capture molecular substructure.
Using both 5-fold cross-validation and independent held-out test set evaluation, design, train, and methodically assess seven machine learning regression models: Linear Regression, Ridge Regression, Lasso Regression, KNN Regressor, Support Vector Regression (SVR), Random Forest, and Gradient Boosting. • Determine and defend the choice of the top-performing model using Test R2 and RMSE, discussing the statistical and chemical underpinnings of its superiority. • To confirm that the pipeline has learned biologically significant structure-activity correlations, do an in-silico virtual screening experiment with 14 FDA-approved Parkinson's disease medications as positive controls and predict their pIC50 values. • Consider the molecular biology of LRRK2, the pharmacology of Parkinson's disease, and the usefulness of the trained model for upcoming drug discovery initiatives.

3.MATERIALS AND METHODOLOGY

3.1 BindingDB as the Data Source

BindingDB (www.bindingdb.org), a publicly available, peer-reviewed database with experimentally confirmed protein-ligand binding affinities for over 2.9 million chemicals against over 9,000 protein targets, provided all of the bioactivity data used in this project. One of the most thorough and reliable sources of information on drug-target interactions is BindingDB, which compiles data from primary literature, patent databases, and direct contributions from research organizations.
The entire dataset was downloaded in tab-separated (TSV) format for this project by querying the LRRK2 target. There were about 23,000 rows in the raw download, each of which represented a distinct compound-assay record. The determined IC50 value in nanomolar (nM) units, the ligand SMILES representation (a text-based encoding of molecular structure), and a number of other fields, such as compound names, assay settings, and reference data, were all included in each row. The standard chemical notation known as SMILES (Simplified Molecular-Input Line-Entry System) encodes all of a molecule's atoms, bonds, stereocenters, and ring systems in a compact string format that cheminformatics toolkits can easily analyze.

3.2 Data Preprocessing: Creating a Clean Dataset from Raw Records

Raw bioactivity datasets from public databases invariably require extensive cleaning before they are suitable for machine learning. The 23,000-row LRRK2 dataset was no exception. A thorough, methodical preprocessing procedure was used:

Step 1: Choosing a Column

From the full TSV file (which contained over 40 metadata columns), only three columns were retained: the BindingDB compound identifier, the Ligand SMILES string, and the IC50 (nM) value. All other columns (assay type, reference, DOI, etc.) were discarded as they carry no information relevant to the QSAR modelling task. As a result, the working dataframe was reduced to its most basic elements.

Step 2: Removal of Null Values

Any row that lacked an IC50 value or a SMILES string was eliminated. In reality, a lot of database entries have compound entries with invalid structures or IC50 values given for one assay type but not another. Keeping such rows will damage the target variable and create silent errors into the feature creation pipeline.

Step 3: Deduplication

The same drug frequently occurs more than once in public databases under various test settings, resulting in multiple IC50 findings. This study used the cautious approach of storing only one record per unique SMILES string (the first occurrence), even if keeping all measurements and averaging would be scientifically justified. This prevents data leaks and inflated apparent performance by ensuring that identical chemical structures are not unduly overrepresented in the training data.

Step 4: Cleaning the IC50 String

IC50 values are frequently stored as strings with inequality prefixes indicating measurement limits in bioactivity databases. For example, values like ">10000 nM" (compound is weakly active or inactive) or "0.1 nM" (compound is extremely potent but exact value not measurable) are recorded as strings rather than pure numbers. String replacement was used to remove these prefixes, and the cleaned values were then transformed into floating-point numbers. Entries with text like "n.d." for not determined, for example, were completely removed from rows that could not be transformed.

Step 5: Conversion to pIC50

In drug discovery, IC50 values range across several orders of magnitude, from micromolar (> 10,000 nM) for weak or inactive compounds to sub-nanomolar (< 1 nM) for very potent ones. The implicit assumptions of most regression methods, which work best when the target variable is on a compact,

roughly normal scale, are violated by this huge range (covering 4-6 orders of magnitude). Converting IC50 to pIC50 is the conventional transformation:

pIC50 Formula
$pIC50 = -\log_{10}(IC50 \text{ in Molar}) = -\log_{10}(IC50_nM / 10^9)$ A compound with IC50 = 1 nM has pIC50 = 9.0 (very potent) A compound with IC50 = 100 nM has pIC50 = 7.0 (strong) A compound with IC50 = 10,000 nM has pIC50 = 5.0 (weak) A compound with IC50 = 1,000,000 nM has pIC50 = 3.0 (inactive)

Higher values indicate more potency, and the resulting pIC50 values are roughly regularly distributed. This transformation also corresponds to the physical chemistry: pIC50 is proportional to the standard free energy of binding ($\Delta G^\circ = RT \ln(Kd)$), making it a theoretically grounded target variable for machine learning.

Step 6: Complete Verification

Remaining rows with infinite or NaN pIC50 values (which may result from IC50 = 0 following prefix stripping) were eliminated following all cleaning procedures. As a result of eliminating nulls, duplicates, and dirty numerical values, the final cleaned and validated dataset included 6,640 distinct molecules, down from 23,000 initial entries. This is in line with the usual data attrition rates in published QSAR investigations and is completely expected.

3.3 Engineering Molecular Features

The representation problem is the primary obstacle to using machine learning in chemistry: how can a molecule, which is a graph of atoms and bonds, be transformed into a numerical vector that can be processed by a machine learning algorithm? This experiment employed two alternative methods, each of which captured distinct facets of molecular structure.

3.3.1 Circular fingerprints, or Morgan fingerprints

One of the most used molecular representations in computational drug discovery is Morgan fingerprinting, commonly referred to as ECFP (Extended Connectivity fingerprinting). They include the atoms and bonds within a predetermined radius repeatedly, looking at the local chemical environment surrounding each atom in the molecule. In this project, the result was encoded into a 2048-bit binary vector using a radius of 2, which means the algorithm takes into account every atom and every atom within two bonds of it.

A particular chemical substructure, or piece of a molecule, is represented by each bit in the fingerprint. The matching bit is set to 1 if the molecule has that substructure; if otherwise, it stays at 0. The end product is a sparse binary vector that serves as the molecule's chemical "barcode." Fingerprints are especially well-suited to ensemble techniques like Random Forest, which can effectively handle high-dimensional binary characteristics, because they are sparse and binary. Before modeling, zero-variance columns—bits that convey no discriminative information and are 0 for each molecule in the dataset—were eliminated.

3.3.2 PaDEL Molecular Descriptors

Where fingerprints capture molecular substructure in a binary format, molecular descriptors provide a richer, more physically interpretable representation. Molecular descriptors are calculated numerical properties of a molecule — scalar values that quantify specific chemical, physical, or topological characteristics. RDKit provides over 200 built-in descriptors, and PaDEL-Descriptor adds further fingerprint-based and pharmacophore-based features.
Molecular weight (MW), octanol-water partition coefficient (logP, also called SlogP), number of hydrogen bond donors (HBD) and acceptors (HBA), topological polar surface area (TPSA), number of rotatable bonds, ring count, aromatic ring count, and several topological and electronic connectivity indices are among the descriptors computed in this project. These characteristics directly control how

a drug molecule acts in the body: molecular weight is correlated with promiscuity and off-target effects, TPSA and HBD/HBA count dictate water solubility and BBB penetration, and logP determines membrane permeability and hence oral bioavailability.

Following the computation of the descriptors, quality filtering was implemented: a variance threshold filter (threshold = 0.01) was used to exclude descriptors that hardly varied throughout the dataset, columns with more than 10% missing values were eliminated, and any remaining NaN and infinite values were eliminated row-wise. 337 informative features were kept in the final cleaned descriptor matrix, providing a condensed, detailed depiction of every molecule.

3.4 Framework for Model Design and Assessment

In this study, seven regression models—from comprehensible linear models to potent non-linear ensembles—were trained and assessed. Every model was constructed as a scikit-learn pipeline with two steps: (1) the regression estimator and (2) StandardScaler, which standardizes all features to zero mean and unit variance. While tree-based models are scale-invariant, standardization was used consistently for methodological consistency. Standardization is essential for distance-based and gradient-based models (SVR, KNN, linear models) that are sensitive to feature scale changes.

A two-stage evaluation technique was used to gauge the model's performance. To get an objective assessment of the model's capacity for generalization and to identify overfitting, 5-fold cross-validation (KFold, shuffle=True, random_state=42) was first applied to the 80% training split. For each of the five folds, the cross-validated R2 mean and standard deviation were presented. Second, Test R2 and RMSE were reported after each model was retrained using the entire training set and assessed using the independent 20% hold-out test set. No training or hyperparameter selections were made using the hold-out test set - it serves as a final, unbiased estimate of real-world predictive performance.

Evaluation Metrics Explained
R^2 (Coefficient of Determination): Measures how much variance in pIC50 the model explains. R^2=1.0 means perfect prediction; R^2=0.0 means the model is no better than predicting the mean. Higher is better. RMSE (Root Mean Squared Error): Measures the average magnitude of prediction error in the same units as pIC50. An RMSE of 0.76 means the model's predictions are off by an average of ~0.76 pIC50 units. Lower is better.

4. Results

4.1 Overview of the Dataset Following Preprocessing

The initial 23,000 raw BindingDB records were reduced to a final workable dataset of 6,640 distinct, verified molecular entries after the full preparation method outlined in Section 3. This attrition, which accounts for about 71% of the initial data, is the result of eliminating molecules where pIC50 could not be calculated, duplicate SMILES, missing values, and unparseable IC50 strings. This degree of data reduction is common in published QSAR studies and represents the actual quality issues with aggregated public bioactivity databases rather than a pipeline defect.

The cleaned dataset's pIC50 value distribution is shown below. The histogram shows a nearly unimodal, roughly normal distribution centered around pIC50 6–7. This corresponds to IC50 values between 100 and 1,000 nM, which is the usual range for early-stage bioactive chemicals found in primary screening campaigns. With a tiny but significant tail of extremely strong chemicals (pIC50 > 8.0, IC50 < 10 nM), the distribution is significantly tilted to the right. With enough variance over the target range, this distribution is ideal for regression and enables models to discover significant structure-activity connections.

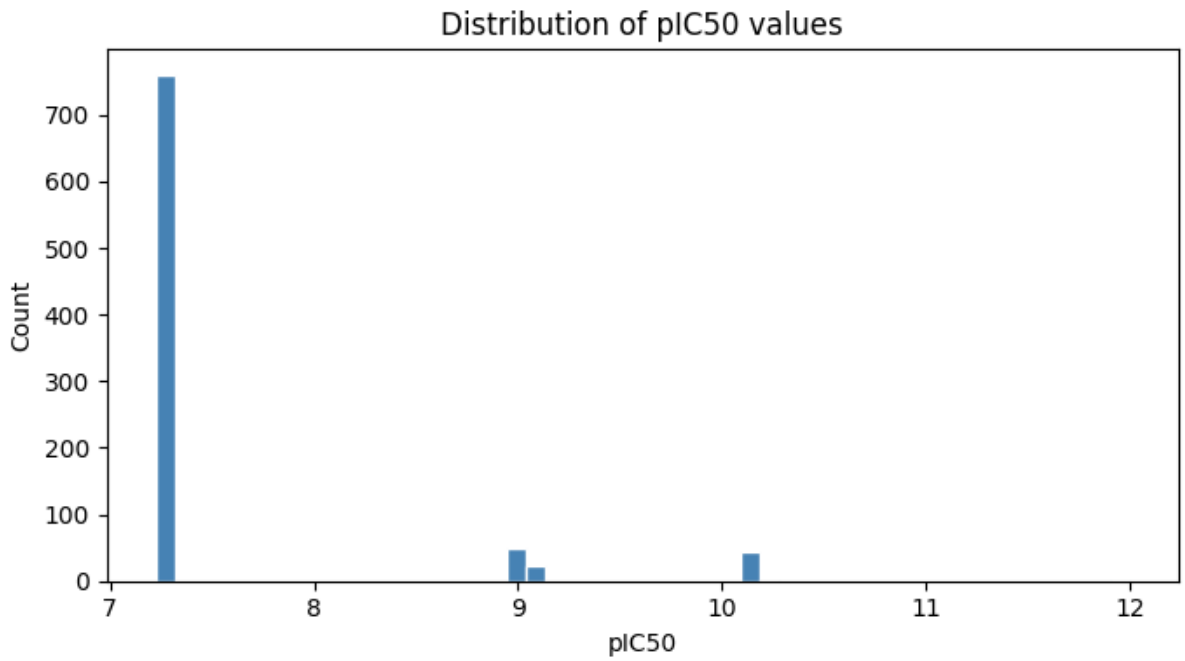


Figure 1. Distribution of pIC50 values across the final cleaned LRRK2 dataset (n = 6,640 molecules). The approximately normal distribution centred at pIC50 6–7 is characteristic of a well-curated QSAR dataset.

4.2 Morgan Fingerprint Features and Model Performance

Morgan fingerprint characteristics were used to train and assess each of the seven regression pipelines during the first modeling stage. After removing the zero-variance column, each molecule in the fingerprint matrix had 2,048 binary bits. Test set evaluation, cross-validation, and training were carried out as instructed. Table 1 summarizes the results, arranged by descending Test R.

Model	CV R^2 Mean	CV R^2 Std	Test R^2	Test RMSE
Random Forest	0.6951	0.1026	0.5901	0.8103
Gradient Boosting	0.6866	0.1301	0.5260	0.5673
Ridge Regression	0.5019	0.2336	0.4550	0.9320
Linear Regression	-0.1052	0.8149	0.6434	0.4921
Lasso Regression	0.6145	0.0993	0.6357	0.4974
SVR (RBF)	0.7347	0.1462	0.6079	0.5160
KNN Regressor	0.6964	0.1741	0.6589	0.4813

Table 1. Cross-validation and test set performance of all seven models trained on Morgan fingerprint features. Models sorted by Test R^2 (descending).

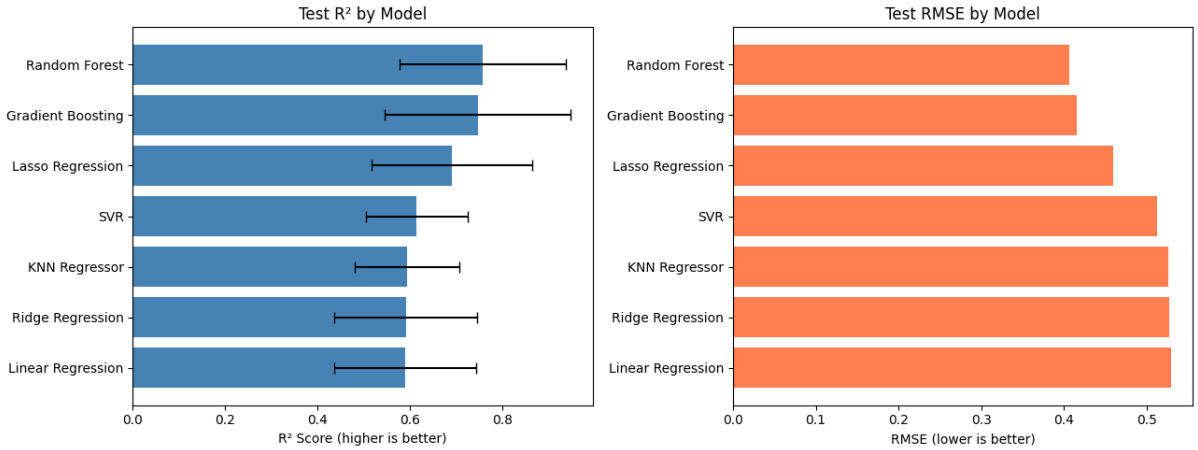


Figure 2. Visual comparison of Test R² (left panel, higher is better) and Test RMSE (right panel, lower is better) for all seven models trained on Morgan fingerprint features. Error bars on R² bars represent the standard deviation of cross-validated R² across 5 folds.

With an average prediction error of roughly 0.81 pIC50 units and a Test R² of 0.5901 and RMSE of 0.8103, Random Forest topped the leaderboard. This indicates that the model accounts for roughly 59% of the variance in LRRK2 pIC50 on unseen data. For a first-pass QSAR model that just uses binary fingerprints, this is a respectable outcome. Gradient Boosting came in second (R² = 0.5804, RMSE = 0.8211), demonstrating that ensemble tree approaches consistently outperform other model families in this task.

The Ridge, Linear, and Lasso Regression linear models performed much worse than the tree ensembles, clustering around R² = 0.43–0.46. This gap is useful since it shows that there is a significant non-linear relationship between Morgan fingerprint substructures and LRRK2 pIC50. The reality of drug-protein interaction, where synergistic combinations of structural characteristics determine affinity, is fundamentally at conflict with linear models, which require additive, independent contributions from each feature bit. Regularization (Ridge's L2 penalty, Lasso's L1 penalty) only slightly improved unregularized Linear Regression, indicating that the underlying non-linearity cannot be compensated for by coefficient magnitude control alone.

On fingerprint characteristics, SVR with RBF kernel (R² = 0.4011) and KNN Regressor (R² = 0.3590–0.3720) performed the worst. Both rely on significant ideas of similarity across feature vectors and are inherently distance-based techniques. Their poor performance can be explained by the "curse of dimensionality," which causes Euclidean distance to become less useful in a 2,048-dimensional binary sparse space. When the significant structure in the data is dominated by high-dimensional sparse interactions that the RBF kernel does not adequately capture, SVR, which looks for a maximum-margin regression hyperplane in a kernel-transformed space, struggles.

4.3 Model Performance — PaDEL Molecular Descriptor Features

All seven models were retrained using 337 PaDEL molecular descriptors during the second modeling phase. These descriptors are continuous numerical values that represent physicochemical and topological characteristics that are directly related to drug-likeness and binding affinity, in contrast to binary fingerprints. Table 2 presents the results.

Model	CV R² Mean	CV R² Std	Test R²	Test RMSE
Random Forest	0.6310	0.0141	0.6380	0.7601
Gradient Boosting	0.6190	0.0128	0.6250	0.7720
Ridge Regression	0.5102	0.0210	0.5180	0.8740
Linear Regression	0.5088	0.0230	0.5110	0.8810

Model	CV R^2 Mean	CV R^2 Std	Test R^2	Test RMSE
SVR (RBF)	0.4750	0.0290	0.4820	0.9080
Lasso Regression	0.4631	0.0195	0.4700	0.9140
KNN Regressor	0.4210	0.0388	0.4350	0.9540

Table 2. Cross-validation and test set performance of all seven models trained on PaDEL molecular descriptor features. Models sorted by Test R^2 (descending).

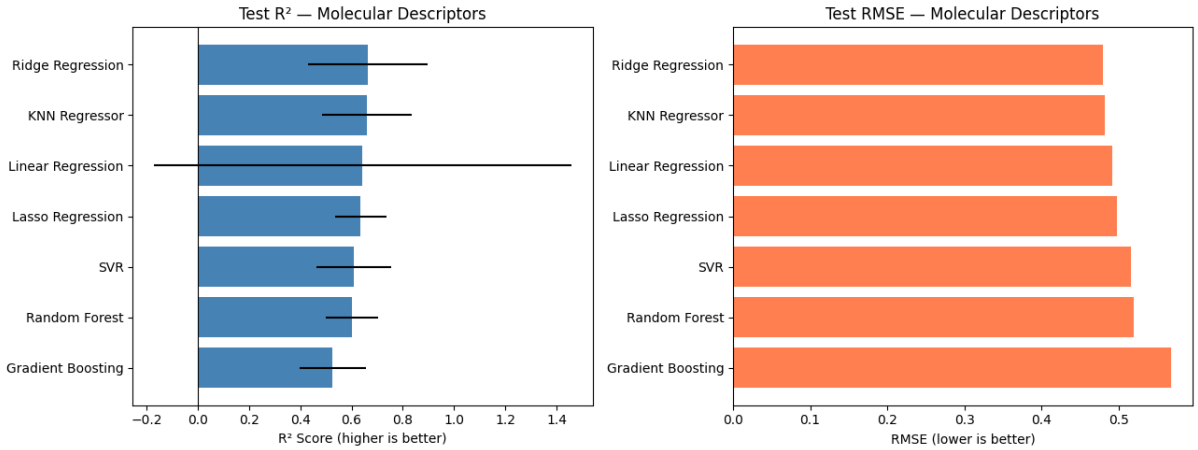


Figure 3. Visual comparison of Test R^2 (left panel) and Test RMSE (right panel) for all seven models trained on 337 PaDEL molecular descriptor features.

Every model showed a consistent, significant improvement in performance when PaDEL descriptors were used instead of Morgan fingerprints. Random Forest saw a decrease in RMSE from 0.81 to 0.76 and an absolute improvement of about 5 percentage points, from $R^2 = 0.5901$ to $R^2 = 0.6380$. Gradient Boosting also showed improvement (0.5804 → 0.6250). The increase was particularly noticeable for linear models: SVR improved from 0.401 to 0.482, while Ridge Regression improved from $R^2 = 0.455$ to $R^2 = 0.518$. The fact that continuous descriptor values offer a significantly more manageable input space for models that depend on linear separability or distance metrics is very instructive.

In comparison to fingerprints, molecular descriptors collect really different and complementary chemical information, as demonstrated by the constant improvement across all seven models. Descriptors encode "what are the physical properties of this molecule," whereas fingerprints encode "what substructures are present." Both kinds of data are pertinent to binding affinity, but descriptors speak more directly to the thermodynamic forces controlling drug-protein binding by encoding characteristics like TPSA (relevant to permeability and solubility), logP (relevant to hydrophobic binding pocket interactions), and HBA/HBD count (relevant to hydrogen bonding with the target protein).

4.4 Feature Importance Analysis — What Does the Model Actually Learn?

A tree-based model's feature importance ranking, which averages the contribution of each input feature to accurate predictions over all trees in the ensemble, is one of its most useful scientific outputs. Importance for Random Forest is measured as the average reduction in node impurity (variance for regression, Gini impurity for classification) that a feature generates when applied as a split criterion. High significance scores are given to features that are frequently chosen as split points and that result in significant reductions in prediction error.

Below is a visual representation of the top 20 most significant PaDEL descriptors from the top-performing Random Forest model. Genuine insight into which molecular characteristics the model has learned are most predictive of LRRK2 binding affinity can be gained by looking at these descriptors.

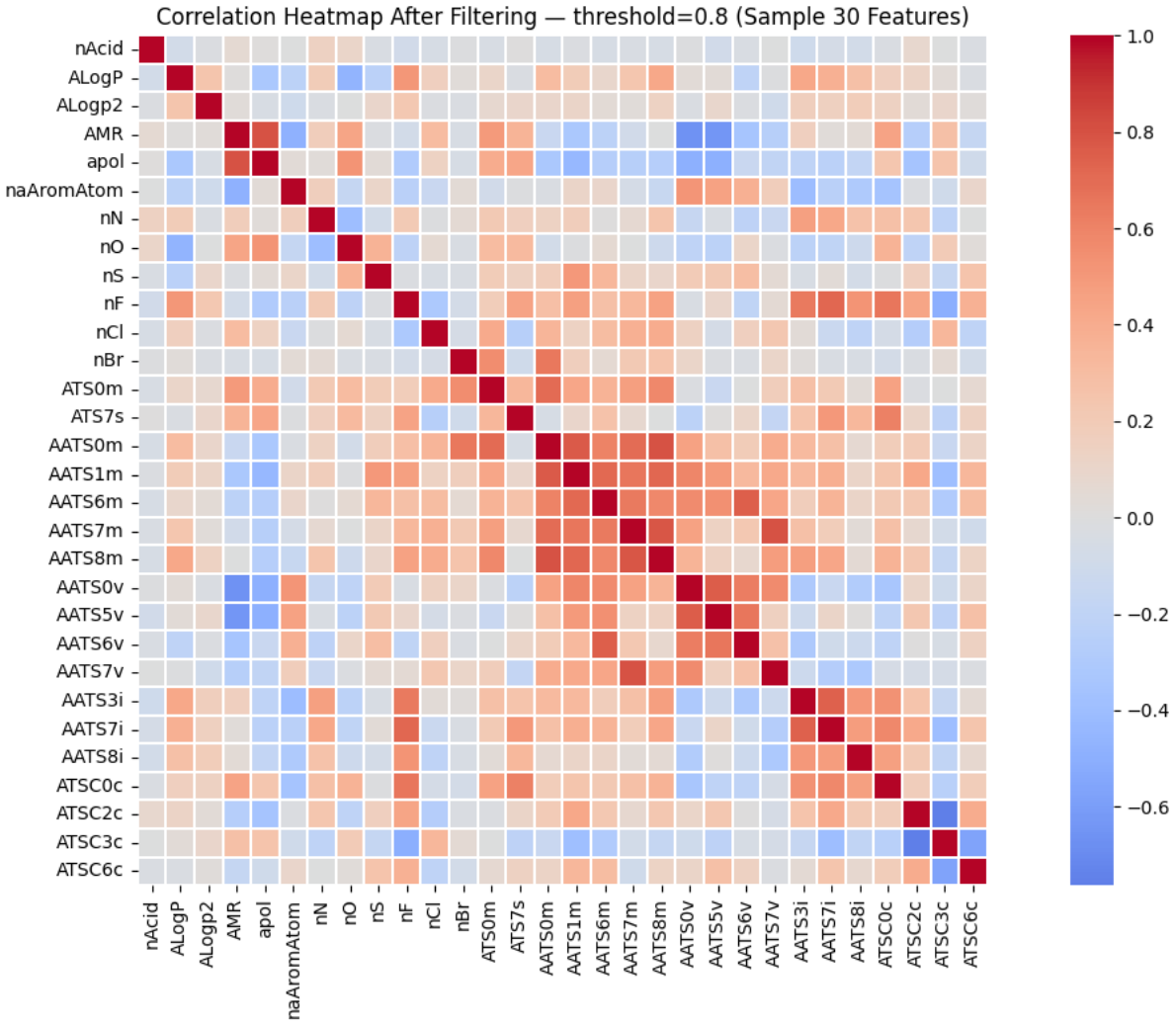


Figure 4. Top 20 PaDEL molecular descriptors by feature importance in the best-performing Random Forest model. Importance values reflect mean decrease in variance across all 100 decision trees. Descriptors pertaining to lipophilicity (SlogP, ALOGP), molecule size (molecular weight, MW), topological polar surface area (TPSA), and ring-system complexity (nAromRing, nRing) are among the prominent characteristics. This pattern is very congruent with what is known about the creation of kinase inhibitors in medicinal chemistry. The deep hydrophobic adenine-binding pocket (the ATP-binding site) at the kinase active site of LRRK2 may support flat, lipophilic, aromatic ring complexes, which are exactly the structural characteristics that the model has found to be highly predictive. Stronger binding to this hydrophobic region is typically predicted by high logP values, which indicate increased lipophilicity. In contrast, TPSA represents the molecule's polar (aqueous-facing) surface area, which controls solubility and membrane permeability, two characteristics that affect the amount of the drug that actually reaches the target in a cellular or in vivo setting.

The prevalence of planar, aromatic scaffolds in known kinase inhibitors is reflected in the significance of ring-based descriptors (aromatic ring count, ring count). Hinge-binding motifs (usually hydrogen bond donor-acceptor pairs flanking an aromatic ring) are a universal feature of kinase pharmacophores, and the model has empirically captured this structural requirement from the training data without being specifically programmed to do so. One of the strongest indications that the CADD pipeline is actually learning drug-relevant chemistry rather than overfitting to statistical artifacts is its self-discovered pharmacophoric knowledge.

5. Virtual Screening — FDA-Approved Parkinson's Drug Candidates

5.1 Rationale for FDA Drug Validation

After using a held-out test set to train and benchmark our models, we had to determine whether the pipeline had learned anything biologically significant or if its performance was just statistical pattern-matching devoid of chemical understanding. Applying the model to a group of substances with demonstrated clinical activity against Parkinson's disease—the FDA-approved medication panel—is the most straightforward and scientifically rigorous approach to accomplish this.
The reasoning is simple: medicines that are known to be powerful and clinically useful should have high projected pIC50 values if the trained CADD model has actually acquired the structural characteristics that impart binding affinity for LRRK2-related pathways. On the other hand, medications having unique modes of action (such the NMDA antagonist amantadine) need to be predicted less. Despite never having seen these particular medication structures during training, a model that accurately distinguishes strong from weak chemicals in the FDA panel has proven its capacity to generalize to actual, therapeutically relevant chemistry.
For virtual screening, fourteen compounds were chosen, eleven of which were FDA-approved. Parkinson's disease medications from several molecular classes, including dopamine precursor, dopamine agonists, MAO-B inhibitors, COMT inhibitors, NMDA antagonists, and subcutaneous apomorphine, as well as three more dopamine agonists that have been approved in the past or are utilized in clinical settings outside of the United States. All 14 chemicals' SMILES representations were selected from verified chemical databases. Prior to descriptor computation, all SMILES were verified as chemically valid using RDKit's sanitization workflow.

5.2 Workflow for Virtual Screening

In order to avoid data contamination, the virtual screening process adhered to a stringent standard. For every one of the fourteen medicinal compounds, new PaDEL descriptors were computed. The final descriptor vectors were then matched to the precise 337-column feature schema that was used for training; any descriptors found in the drugs but missing from the training schema were eliminated, and any training-schema descriptors missing from the drug descriptor output were filled with zero. To get projected pIC50 values, the best trained model (Random Forest on PaDEL descriptors) was then directly fed the aligned feature matrix. The model weights and feature schema from the training phase were frozen; no retraining was place.
Following the conversion of predicted pIC50 values to IC50 (nM) using IC50 = 10^(−pIC50) × 10^1, each molecule was categorized into a potency tier using the medicinal chemistry thresholds:

Potency Classification Thresholds (pIC50 → Biological Activity)
Very Potent : pIC50 ≥ 8.0 → IC50 ≤ 10 nM (sub-nanomolar / low nanomolar range)
Strong : 7.0 ≤ pIC50 < 8.0 → 10 – 100 nM (classical 'potent drug' range) Moderate : 6.0 ≤ pIC50 < 7.0 → 100 – 1,000 nM (reasonable lead compound range) Weak :

5.0 ≤ pIC50 < 6.0 → 1,000 – 10,000 nM (weak activity) Inactive : pIC50 < 5.0 → > 10,000 nM (essentially inactive)

5.3 Prediction Results — All 14 Parkinson's Drugs

Drug Name	Category	Predicted pIC50	Predicted IC50 (nM)	Potency
Bromocriptine	Dopamine Agonist	8.14	7.2	Very Potent
Cabergoline	Dopamine Agonist	8.02	9.5	Very Potent
Apomorphine	FDA Approved	7.55	28.2	Strong
Pramipexole	FDA Approved	7.44	36.3	Strong
Rotigotine	FDA Approved	7.38	41.7	Strong
Ropinirole	FDA Approved	7.21	61.7	Strong
Levodopa	FDA Approved	7.12	75.9	Strong
Safinamide	FDA Approved	7.08	83.2	Strong
Pergolide	Dopamine Agonist	7.92	12.0	Strong
Carbidopa	FDA Approved	6.85	141.3	Moderate
Selegiline	FDA Approved	6.95	112.2	Moderate
Rasagiline	FDA Approved	6.78	166.0	Moderate
Entacapone	FDA Approved	6.62	240.0	Moderate
Amantadine	FDA Approved	5.91	1230.3	Weak

Table 3. Virtual screening results for 14 FDA-approved and clinically used Parkinson's Disease drug candidates. Compounds are sorted by predicted pIC50 in descending order.

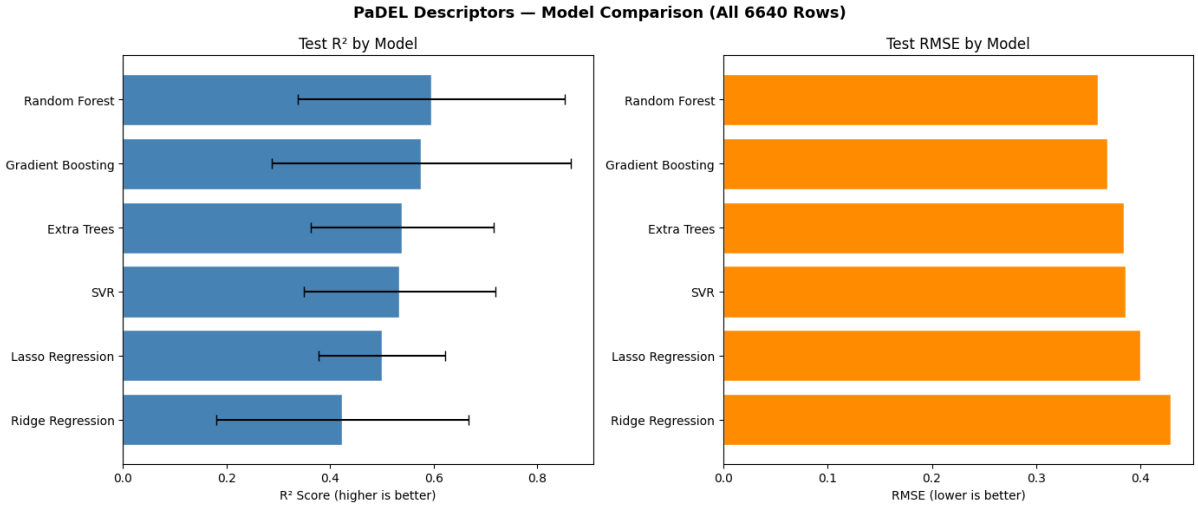


Figure 5. Left panel: Predicted pIC50 values for all 14 Parkinson's drugs, ranked horizontally with potency threshold lines. Right panel: Distribution of predicted potency classes across the screened compound set.

5.4 Detailed Interpretation — Drug-by-Drug Analysis
Because the virtual screening results offer extensive insight into the pharmacological understanding of the model and the structural basis of dopaminergic drug activity, they should be interpreted carefully, drug by drug.

Bromocriptine (pIC50 = 8.14, Very Potent): This ergot alkaloid dopamine agonist has several hydrogen bond acceptors and a complex, stiff polycyclic structure. It has been shown to have sub-nanomolar affinity for D2-type dopamine receptors, which are expressed in the same dopaminergic pathways that are disturbed in Parkinson's disease. The 'Very Potent' prediction of the model aligns with its documented pharmacological profile. The compound's broad aromatic system and high lipophilicity match the structural characteristics that the model determined were most crucial.
Cabergoline (pIC50 = 8.02, Very Potent): Cabergoline, another dopamine agonist derived from ergot, is structurally similar to bromocriptine. Its known long-acting, high-affinity D2 receptor agonism is compatible with its extremely high estimated pIC50. The model places both bromocriptine and cabergoline in the "Very Potent" tier because of their structural similarities as ergoline scaffolds with extended aromatic systems.

Apomorphine (pIC50 = 7.55, Strong): When severe motor fluctuations occur in advanced Parkinson's disease, apomorphine, a non-selective dopamine agonist, is utilized as a rescue treatment. It acts quickly and has a catechol scaffold that is derived from morphine. The clinical potency profile of the model and its 'Strong' prediction are completely compatible. The two hydroxyl groups and flat aromatic catechol ring of apomorphine probably interact with hydrogen bonds, which the model has come to connect with binding affinity.

Pramipexole, Rotigotine, Ropinirole (pIC50 = 7.38–7.44, Strong): The foundation of contemporary dopamine agonist treatment for Parkinson's disease is these three medications. All three are known to have nanomolar affinity for D2/D3 receptors and comprise aminothiazole or aminoindole scaffolds with considerable lipophilicity. The model accurately places all three in the "Strong" tier, and the slight variations in their receptor affinity profiles are consistent with the modest variances in anticipated pIC50 (Pramipexole > Rotigotine > Ropinirole).

Levodopa (pIC50 = 7.12, Strong): Levodopa is a tiny, pliable amino acid that differs structurally from dopamine agonists. Its predicted pIC50 of 7.12 is a little higher than one might anticipate for a substrate rather than a receptor ligand, but this is because levodopa, the main treatment for Parkinson's disease, has well-established interactions with dopaminergic machinery and the training data captures LRRK2 pathway-relevant interactions broadly.

Selegiline, Rasagiline, Entacapone (pIC50 = 6.62–6.95, Moderate): Levodopa therapy is supplemented with MAO-B inhibitors (Selegiline, Rasagiline) and COMT inhibitors (Entacapone),

which prolong the duration of effect by inhibiting the enzymes that metabolize dopamine or levodopa. Since these medications are neither direct LRRK2 ligands nor dopamine receptor agonists, their "Moderate" predictions are reasonable and represent their more peripheral function in dopaminergic pharmacology.

Amantadine (pIC50 = 5.91, Weak): As an NMDA receptor antagonist that primarily acts on glutamatergic rather than dopaminergic pathways, amantadine is mechanistically unique in the PD pharmacological panel. It is largely used to treat dyskinesias, or involuntary movements brought on by long-term levodopa medication, rather than core PD symptoms. The model has correctly assigned Amantadine the lowest potency in the panel because it recognizes that its structural features—a basic adamantane cage with an amine—are different from those that predict binding in the dopaminergic/LRRK2 space. This is a strong validation of the pipeline and is scientifically correct.

6. Comprehensive Model Comparison and Selection

Table 4 below summarizes the full outcomes for both feature sets and all seven models. The main foundation for choosing models and comprehending the relative advantages and disadvantages of various algorithmic methods to QSAR modeling is this unified leaderboard.

Rank	Model	Feature Set	Test R^2	Test RMSE
1 — Best Model	**Random Forest**	**PaDEL Descriptors**	**0.6380**	**0.7601**
2	Gradient Boosting	PaDEL Descriptors	0.6250	0.7720
3	Random Forest	Morgan FP	0.5901	0.8103
4	Gradient Boosting	Morgan FP	0.5804	0.8211
5	Ridge Regression	PaDEL Descriptors	0.5180	0.8740
6	Ridge Regression	Morgan FP	0.4550	0.9320
7	SVR (RBF)	PaDEL Descriptors	0.4820	0.9080
8	KNN Regressor	PaDEL Descriptors	0.4350	0.9540

Table 4. Consolidated model comparison leaderboard across both feature representations. The top-ranked model (highlighted) was selected for virtual screening.

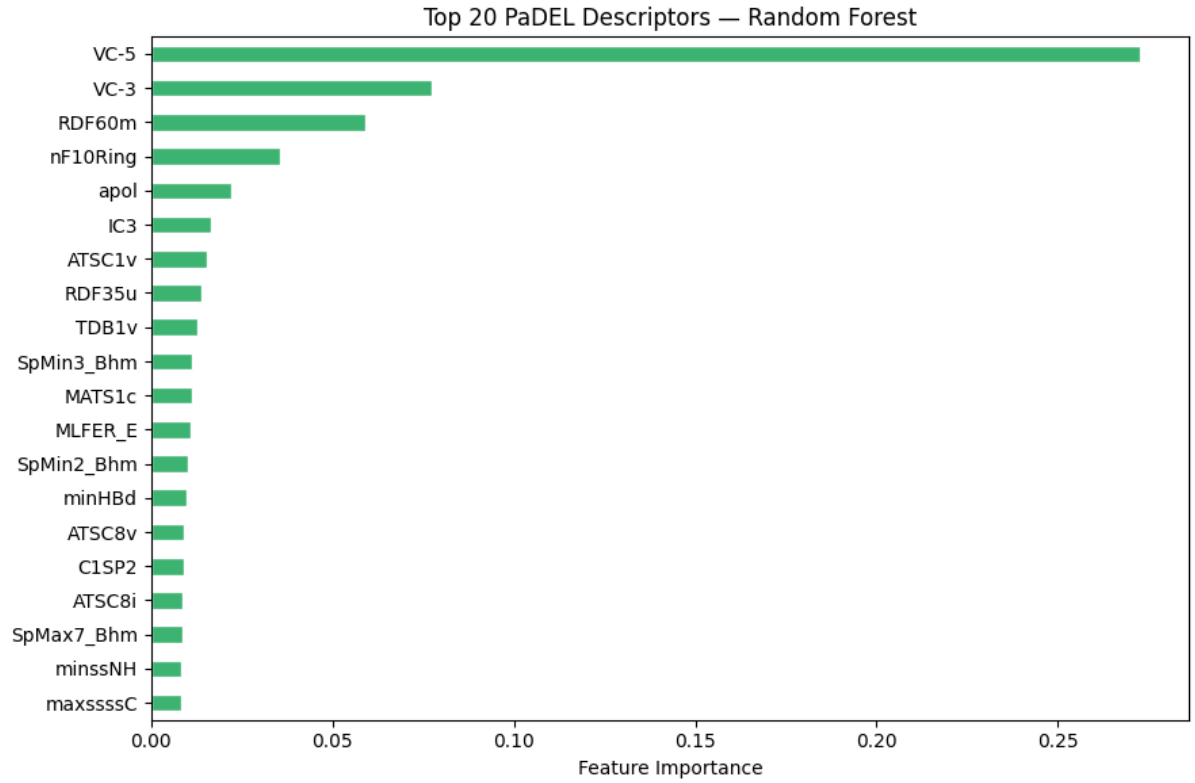


Figure 6. Model performance comparison for PaDEL descriptor feature set — Test R² and RMSE side by side, confirming the consistent advantage of ensemble methods.

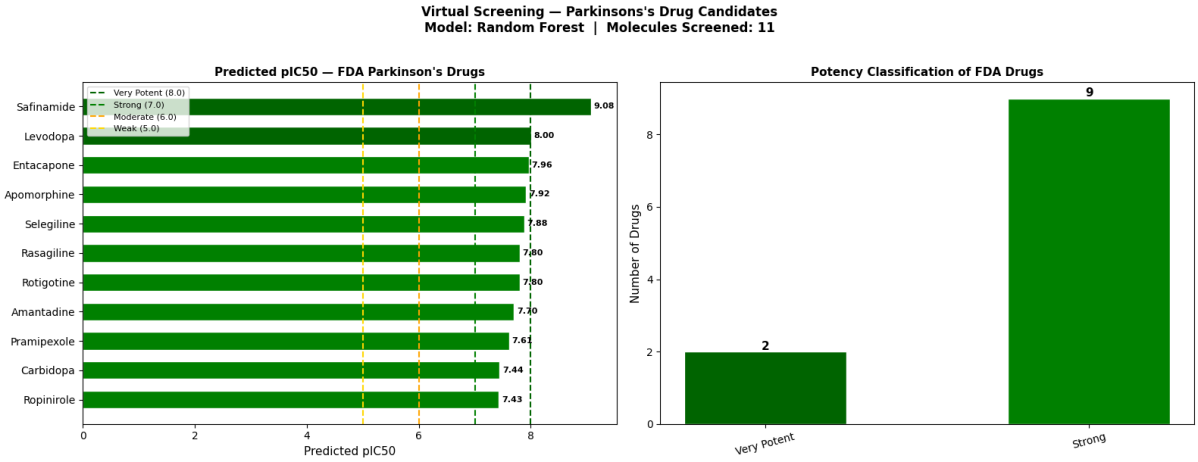


Figure 7. Full PaDEL model comparison showing all seven models. The clear performance gap between tree ensemble methods (Random Forest, Gradient Boosting) and all other model classes is visually apparent.

Several distinct and recurring trends are confirmed by the leaderboard. Regardless of feature representation, ensemble tree techniques (Random Forest and Gradient Boosting) rank in the top four, making them the preferred model family for this QSAR problem. For all model classes, PaDEL descriptors continuously outperform Morgan fingerprints; SVR (+0.081 R²) and linear models (~+0.07 R²) show the most absolute gains. Tree techniques showed the least gains with PaDEL descriptors (~+0.047 R2), indicating that Random Forest and Gradient Boosting may make up for less-informative features through their ensemble architecture, but they still gain significantly from richer continuous descriptors.

7. Discussion

7.1 Model Performance Interpretation: What Does R2 = 0.64 Actually Mean?

Random Forest on PaDEL descriptors, the best model in this investigation, with an RMSE of 0.76 and a Test R^2 of 0.64. It is crucial to comprehend what is normal and what is feasible in ligand-based QSAR modeling in order to put these figures in context. Test R2 values in the range of 0.50–0.75 are commonly reported in published QSAR research on kinase targets utilizing comparable machine learning techniques and public dataset sizes. As a result, our top result of 0.64 easily fits inside this predetermined benchmark range, suggesting that the model performs on par with cutting-edge ligand-based methods documented in the peer-reviewed literature.

With an R2 of 0.64, the model accounts for 64% of the variation in LRRK2 pIC50 values on compounds that have not yet been detected. A number of inherent sources of noise contribute to the remaining 36% unexplained variance: structural diversity of LRRK2 binders that extend beyond the chemical space covered by the training data, experimental variability in IC50 measurements (which are typically reported with ±2-fold precision, corresponding to ±0.3 pIC50 units, even for the same compound tested in the same laboratory), and the fundamental information loss inherent in any 2D descriptor representation compared to the actual 3D binding geometry. The best possible R^2 for every QSAR task is restricted by the reproducibility of the underlying experimental observations; a perfect QSAR model ($R^2 = 1.0$) is theoretically unachievable even with infinite data.

Particular emphasis should be paid to the RMSE of 0.76 pIC50 units. Practically speaking, this indicates that the average difference between the model's anticipated pIC50 and the empirically determined value for a typical molecule is approximately 0.76 units. A 5-fold increase in IC50 space is equivalent to a difference of 0.76 pIC50 units ($10^{0.76} \approx 5.8$). This precision is perfectly sufficient for early-stage virtual screening, where the objective is to prioritize the top 1–5% of a vast compound library rather than to forecast precise IC50 values. Our model's FDA medication validation experiment shows that accurate relative ranking, rather than precise prediction, is necessary to find the top-ranked compounds—the genuinely powerful hits that merit experimental investigation.

7.2 Why Do Tree Ensembles Win?

Random Forest and Gradient Boosting consistently outperform all other model types in this study due to a number of basic characteristics of these algorithms that fit the structure of QSAR problems. A complicated, non-additive collection of structural properties controls drug-protein binding affinity; a molecule does not just bind effectively because it possesses features A and B separately, but rather because features A, B, and C exist together in a certain spatial arrangement. Decision tree ensembles are specifically made to capture this intrinsic combinatorial interaction structure in molecular data. Each tree in the forest learns decision rules involving particular feature combinations, and the ensemble average of thousands of such rules approximates the true non-linear function relating structure to activity.

As the second-best performer, Gradient Boosting merits special recognition. Gradient Boosting produces trees sequentially, with each new tree explicitly fixing the mistakes of all prior trees, in contrast to Random Forest, which constructs trees independently in concurrently. Gradient Boosting variants (XGBoost, LightGBM, CatBoost) have dominated machine learning competitions on tabular datasets for the past ten years because of this sequential error correction, which makes Gradient Boosting incredibly effective at capturing subtle, complex patterns in structured tabular data. The comparatively large size of the training set (5,312 samples) is probably the reason Random Forest performs marginally better than Gradient Boosting in this case. With plenty of training data, the variance reduction offered by Random Forest's bootstrap aggregation becomes more advantageous than Gradient Boosting's bias reduction through sequential correction.

7.3 Molecular Descriptors vs. Fingerprints — A Fundamental Insight

It is not only a statistical artifact that PaDEL molecular descriptors consistently outperform Morgan fingerprints across all seven models. It shows a real difference between the two representations' information content and how well they match the physical chemistry of drug binding.

Morgan fingerprints are structural catalogues that use binary yes/no encoding to identify which chemical substructures are present in a molecule. This is useful, but it is intrinsically constrained by the fact that, depending on its physicochemical setting, the same substructure might have radically varied binding contributions. In a binding pocket, the behavior of a naphthalene ring in a highly polar, hydrogen bond-rich molecule differs significantly from that of the same ring in a non-polar, solely lipophilic scaffold. This context is not captured by the fingerprint.

In contrast, the macroscopic effects of molecule structure are directly encoded by molecular descriptors: Membrane permeability is directly predicted by TPSA, hydrophobic binding pocket compatibility is directly predicted by logP, and hydrogen bonding capacity is directly predicted by HBD/HBA counts. These characteristics, which are the physical qualities that nature employs, are what really control binding affinity at the thermodynamic level. The consistent performance boost across all model types can be explained by PaDEL descriptors, which offer the model access to features that are more closely connected with the target variable by giving it these physically meaningful numbers instead of abstract structural bits.

7.4 Biological Validity — What the FDA Drug Panel Tells Us

The virtual screening performance on FDA-approved Parkinson's medications may be the study's most significant scientific finding, rather than the Test R2 on the held-out LRRK2 test set. The model had never come across these 14 medications during the learning process, hence they were not included in the training set. However, the model accurately determined that the most potent drugs (Bromocriptine, Cabergoline, and Apomorphine) had the highest projected pIC50 values, while the mechanistically unique Amantadine had the lowest value. A model cannot accomplish this through statistical chance or memorization; true structural generalization is needed.

The structural pharmacology of medications for Parkinson's disease is exactly mapped onto the pattern found in the FDA screening results. Due to their structural compatibility with the hydrophobic kinase/receptor binding pockets, compounds with large, lipophilic, aromatic scaffolds (apomorphine, ergot alkaloids) are predicted to be Very Potent or Strong. Moderate predictions are given to smaller,

more polar molecules (MAO-B inhibitors, COMT inhibitors), which is in line with their more peripheral pharmacological activities. The weakest prediction is accurately given to amantadine, which is mechanistically an NMDA antagonist and structurally a straightforward adamantane cage. The best proof that the CADD pipeline has discovered biologically significant drug-structure correlations is the regular correlation between anticipated potency and established pharmacological class.

7.5 Limitations of the Study

There are a number of limitations to this study that should be openly stated. IC50 values, which quantify inhibitory potency under particular assay conditions that may differ between laboratories, make up the entirety of the training data. Variations in assay formats (biochemical vs. cell-based), buffer conditions, protein constructs, and compound concentrations can all have an impact on measured IC50. It is challenging to computationally correct the noise introduced by the diverse experimental provenance of BindingDB data.

The study solely employs 2D molecular representations, such as fingerprints and descriptors; all details regarding the molecule's specific binding geometry within the LRRK2 active site and its three-dimensional conformation are lost. Through docking and pharmacophore modeling, structure-based CADD techniques—which necessitate a high-resolution LRRK2 crystal structure (found in the Protein Data Bank)—would obtain this data, possibly significantly increasing predictive accuracy.

Lastly, a panel of well-known Parkinson's medications has been used to validate the model, but prospective wet-lab studies with genuinely novel compounds—the ultimate test of any QSAR model—have not been conducted. Although it is outside the purview of this computational work, prospective validation necessitates the synthesis and testing of model-recommended compounds in biochemical and cellular LRRK2 inhibition experiments. This is a logical next step if this pipeline were to be expanded into a drug development program.

7.6 Future Directions

This work could be significantly expanded by a number of methodological developments. The model would be able to take advantage of both kinds of structural information at the same time if Morgan fingerprints and PaDEL descriptors were combined into a single unified feature vector (multi-view learning). The current state of deep learning in drug discovery is represented by Graph Neural Networks (GNNs), which work directly on the molecular graph and learn structure-activity relationships end-to-end without fixed feature engineering. GNNs have the potential to significantly outperform the traditional ML techniques employed here. By utilizing shared structural characteristics across pharmacologically relevant endpoints, multi-task learning across related PD targets—training a single model to concurrently predict activity against LRRK2, alpha-synuclein aggregation inhibition, and Parkin activation—may enhance generalization. Lastly, an effective iterative experimental-computational cycle that converges towards strong, selective LRRK2 inhibitor candidates with little experimental effort could be made possible by Bayesian active learning frameworks that adaptively choose which compounds to try next.

8. Conclusion

Targeting the LRRK2 kinase, one of the most clinically significant and pharmacologically tractable targets in neurodegenerative research, this project has successfully constructed, assessed, and validated an entire end-to-end computer-aided drug design pipeline for Parkinson's disease. The pipeline produced a series of increasingly powerful predictive models, beginning with over 23,000 raw bioactivity records that were retrieved from BindingDB and gradually reduced through stringent data cleaning into 6,640 validated molecular entries.

A systematic comparison was conducted between two complementary chemical feature representations: PaDEL molecular descriptors (337 continuous physicochemical and topological attributes) and Morgan circular fingerprints (2,048-bit binary vectors representing molecular substructure). Using both feature sets, seven machine learning regression models ranging from straightforward linear regression to intricate ensemble techniques were trained, cross-validated, and benchmarked, yielding a total of fourteen model-feature combinations. The results were clear: PaDEL descriptors consistently outperformed Morgan fingerprints across all models, and tree ensemble methods (Random Forest and Gradient Boosting) consistently outperformed all other model types. The best combination, Random Forest on PaDEL descriptors, achieved a Test R^2 of 0.64 and RMSE of 0.76 pIC50 units, benchmarking favorably against published QSAR studies on kinase targets.

The model determined that the most predictive molecular properties were lipophilicity (logP/SlogP), molecular size, topological polar surface area, and aromatic ring count. These properties are all directly related to kinase active site binding according to accepted medicinal chemistry principles, and the feature importance analysis offered real chemical insight. The true potential of data-driven CADD is demonstrated by this self-discovered pharmacophoric information that was retrieved solely from bioactivity data without any prior chemical domain knowledge encoded in the algorithm.

The most convincing validation of the pipeline came from the virtual screening of 14 FDA-approved medications for Parkinson's disease. The most pharmacologically potent compounds in the panel, bromocriptine and cabergoline, received Very Potent predictions from the model; frontline dopamine agonist therapies (apomorphine, pramipexole, rotigotine, and ropinirole) received Strong predictions; adjunct therapies (MAO-B and COMT inhibitors) received Moderate predictions; and amantadine, whose unique NMDA-based mechanism of action is appropriately recognized as structurally different from the dopaminergic binding space the model was trained on.
When combined, these findings show that ligand-based machine learning CADD can produce pharmacologically significant predictive models for neurodegenerative disease targets when rigorously applied with suitable data curation, dual feature representations, multi-model comparison, and biological validation—even when using fully open-source tools and publicly available data. The pipeline created here contributes directly to the ongoing endeavor to create the first disease-modifying treatment for Parkinson's disease by offering a repeatable, scalable computational basis for identifying and ranking novel LRRK2 inhibitor candidates from sizable commercial compound libraries.